

Verbal Reasoning Richness, Not Domain Proximity, Governs SLM-to-SLM Guided Reasoning: A Statistically Verified Empirical Study --Manuscript Draft--

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June 26, 2026

Dear Editor-in-Chief:

I am pleased to submit an original research article entitled “Verbal Reasoning Richness, Not Domain Proximity, Governs SLM-to-SLM Guided Reasoning: A Statistically Verified Empirical Study” for consideration for publication in the PLOS One Journal.

We introduce GV(Guided Voting), a two-stage inference framework that combines a LoRA-fine-tuned 3B guide, trained via Structured Plan Generation Fine-Tuning (SPGFT), with a frozen 1–1.5B solver under majority-vote decoding. Across six benchmarks spanning arithmetic, commonsense, science, and reading comprehension, GV improves accuracy on five datasets, achieving gains of up to 24.2 percentage points, 13.5 points on average. A third fine-tuning condition (SVAMP) reveals a monotonic richness ordering: GSM8K FT > SVAMP FT > ASDiv FT on most benchmarks, further corroborating the Verbal Richness Hypothesis). Ablation studies isolate fine-tuning quality as the key factor, and a 2×3 factorial analysis across model architectures and three fine-tuning datasets confirms that richer reasoning traces yield consistently better performance. We further identify two failure modes bias-substitution and confident misdirection and observe consistent improvements in calibration, establishing verbal reasoning richness as a primary determinant of effective inference-time guidance in small language models.

We have no conflicts of interest to report regarding the present study. This manuscript has not been published and is not under consideration for publication elsewhere. We have no conflicts of interest to disclose.

Thank you for your consideration.

Sincerely,

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Verbal Reasoning Richness, Not Domain Proximity, Governs SLM-to-SLM Guided Reasoning: A Statistically Verified Empirical Study

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Abstract

A guide model fine-tuned out-of-domain can outperform one trained in-domain, even on the in-domain test set. We term this the *domain proximity paradox* and show that guide effectiveness is driven by the verbal reasoning richness of its training data rather than topical similarity. We introduce GV(Guided Voting), a two-stage inference framework that combines a LoRA-fine-tuned 3B guide, trained via Structured Plan Generation Fine-Tuning (SPGFT), with a frozen 1–1.5B solver under majority-vote decoding. Across six benchmarks spanning arithmetic, commonsense, science, and reading comprehension, GV improves accuracy on five datasets, achieving gains of up to 24.2 percentage points, 13.5 points on average. A third fine-tuning condition (SVAMP) reveals a monotonic richness ordering: GSM8K FT > SVAMP FT > ASDiv FT on most benchmarks, further corroborating the Verbal Richness Hypothesis). Ablation studies isolate fine-tuning quality as the key factor, and a 2×3 factorial analysis across model architectures and three fine-tuning datasets confirms that richer reasoning traces yield consistently better performance. We further identify two failure modes bias-substitution and confident misdirection and observe consistent improvements in calibration, establishing verbal reasoning richness as a primary determinant of effective inference-time guidance in small language models. All code, random seeds, and reproduction instructions are publicly available at <https://github.com/sadi2003q/Small-Language-Model>.

Author summary

Traditional methods of improving reasoning abilities in resource-limited Small Language Models (SLMs) either come at the high cost of scaling up model parameters or depend heavily on very large, private teacher models. In this paper, we propose a new method to remove these costs, Guided Voting (GV), which reduces costs by using an LoRA-fine-tuned 3B guide and a frozen 1 to 1.5B solver in majority-vote Decoding as an inexpensive alternative to large-scale Language Model oracles. We confirm the Verbal Richness Hypothesis and identify a surprising "Domain Proximity Paradox," demonstrating how an out-of-domain guide with long, descriptive, multi-step reasoning patterns outperforms an in-domain guide with the same topic but relatively shallow descriptions on multiple benchmarks in a 2 × 3 factorial design. Moreover, our method provides an average benefit of 13. Later, we will describe some of the pitfalls of using

this method, such as bias substitution, but we hope that this experiment demonstrates how structural support for SLMs is important for achieving effective inference-time guidance from small models regardless of topic similarity.

Introduction

Motivation and Central Question

Improving reasoning in small language models (SLMs) typically requires either scaling model parameters or distilling knowledge from a substantially larger teacher [9, 18]. Both approaches carry significant overhead: parameter scaling multiplies training and serving costs, while distillation demands large annotated reasoning corpora and repeated teacher inference.

Inference-time guidance offers a lighter alternative: a separately trained model generates a structured reasoning plan that conditions a frozen solver. Prior explorations of this paradigm have operated almost exclusively at the scale of large language models [7, 8, 16]. Recent work on solution-guidance fine-tuning [25] demonstrated that LLM-generated guidance—produced by GPT-4o—can substantially improve SLM reasoning, but this retains a critical dependency on large proprietary models, limiting deployment in resource-constrained settings.

We ask a more fundamental question: *when a fine-tuned SLM replaces the LLM guide, what property of its training data determines how well it generalises?*

Our answer, supported by a 2×3 factorial replication across two model families and two fine-tuning datasets, is precise and counterintuitive: **verbal reasoning richness of the training distribution—not its topical proximity to the target task—governs cross-domain transfer**. An in-domain guide trained on single-sentence, single-operation problems is systematically outperformed by an out-of-domain guide trained on multi-step narrative problems, even when evaluated on the in-domain test set. We term this the *domain proximity paradox*.

We do not compare against an LLM-guided baseline, as Bi et al. [25] have already established that LLM guidance (GPT-4o) improves SLM reasoning; our contribution is to ask what property of SLM fine-tuning data replicates this effect without an LLM oracle.

Scientific Contributions

This paper makes the following contributions, ordered by novelty.

- **The Verbal Richness Hypothesis (VRH).** We propose and empirically validate the claim that the structural diversity and step-depth of guide training plans—not their domain overlap with the target—determine downstream guided reasoning performance. The hypothesis is onalised through the SPGFT procedure, which controls plan structure terion deterministic filter, and falsified in one direction by the domain proximity paradox; and confirmed in its richness-ordering prediction by the SVAMP fine-tuning condition, which sits structurally between ASDiv and GSM8K and performs accordingly.
- **Two novel failure modes.** We formally identify and characterise phbias-substitution failure mode—in which a guide overcorrects itional bias, trading one directional error for its mirror—and first systematic taxonomy of C-case errors (*confident misdirection, partial entity neglect, format failure, residual bias*) in guided SLM reasoning.

- **A predictive four-tier task taxonomy.** From the 2×3 replication, we derive a pre-deployment decision procedure that assigns a new task to one of four tiers based on three measurable solver properties, is architecture-conditional and guide-quality-conditional, with actionable deployment rules at each tier.
- **Rigorous statistical validation.** Six benchmarks, more than 2,300 paired evaluations, McNemar’s test with power analysis, effect sizes (OR, Cohen’s g), bootstrap confidence intervals, and multi-seed validation ($\{0, 1, 42\}$, within $\pm 1.5 pp$) across all five confirmed datasets.
- **A compute-efficient self-bootstrapped pipeline.** GUIDEDVOTE eliminates the LLM oracle dependency entirely. The guide is fine-tuned on its own filtered outputs, not on GPT-4o-generated rationales. Combined inference cost is 30–47% below the naïve ceiling of running the 3B model alone.

Paper Organisation

The Related Work section situates the work within the inference-time reasoning and SLM efficiency literatures. The Method section describes the GUIDEDVOTE pipeline. The SPGFT section details SPGFT. The Experimental Setup and Results sections cover the experimental setup and primary results. The Ablation and Analysis sections present the three-pronged ablation and mechanistic analysis, including the two failure modes. The Replication section reports cross-architecture replication and derives the four-tier taxonomy. The ASDiv section operationalises the VRH through the 2×3 data quality ablation across three fine-tuning datasets. The Fine-Tuning Baseline section compares against task-specific solver fine-tuning. The LLM Oracle section establishes the LLM oracle upper bound and quantifies GUIDEDVOTE’s capture rate. The Decision Procedure section provides the pre-deployment decision procedure. The Reproducibility section documents reproducibility details. The Limitations and Conclusion sections conclude.

Related Work

Chain-of-thought and inference-time reasoning. Wei et al. [29] demonstrated that eliciting intermediate reasoning steps substantially improves multi-step performance in large models; Kojima et al. [13] later showed zero-shot activation is achievable with a simple “Let’s think step by step” instruction. Subsequent work has extended this with least-to-most prompting [32], programme-of-thought [2], and tree-of-thoughts [30]. Our findings show that, for SLMs, unstructured step-by-step prompting is insufficient or actively detrimental on four of five evaluation datasets, motivating a fine-tuned plan-generator in its place.

Self-consistency and majority voting. Wang et al. [28] established that aggregating diverse reasoning paths via majority vote improves accuracy without additional training. We adopt this ensemble as our solver’s decoding strategy and show that a fine-tuned guide systematically shifts the vote distribution toward correct answers—increasing what we term *vote consistency lift*—while an untuned guide can instead produce confident, unanimous *incorrect* majorities.

Knowledge distillation for small models. Ho et al. [9] and Magister et al. [18] transfer reasoning capabilities from large models to smaller ones via chain-of-thought distillation. Our approach is orthogonal: the solver is never fine-tuned; only the guide is trained, on a single general-purpose dataset (GSM8K). This separation keeps deployment lightweight and the guide reusable across solvers without retraining.

Solution-guidance fine-tuning. Recent work [25] has explored fine-tuning models on structured solution guidance to improve inference-time reasoning, demonstrating gains on arithmetic and multi-step tasks. Our work extends this paradigm to the SLM-to-SLM regime, introduces a frozen-solver architecture that avoids solver retraining, and—critically— provides the first systematic study of how fine-tuning *data quality* governs transferability: the verbal richness hypothesis and the domain proximity paradox are absent from prior solution-guidance work.

Model collaboration pipelines. Li et al. [16] and Du et al. [7] study multi-agent reasoning at LLM scale; Fu et al. [8] decompose reasoning into specialised roles. We bring this paradigm into the SLM regime and provide rigorous paired statistical evaluation absent from most prior collaborative-agent work.

Calibration and confidence. Desai and Durrett [6] and Kadavath et al. [11] examine whether model confidence reflects true accuracy. We measure ECE [21] before and after guidance, finding universal improvement across all six datasets, including the accuracy-null retrieval case.

Position bias. Systematic positional preferences in MCQ tasks have been documented across model families [12, 23, 31]. We identify the bias-substitution failure mode—a previously undescribed failure in guided MCQ systems—and show that debiasing and content-grounded reasoning are separable, independently necessary components of accuracy gain.

Mixture-of-expert SLM routing. SLM-MUX [33] routes inputs across multiple specialised SLMs via a learned multiplexer, optimising for task-specific expert assignment. Our approach differs fundamentally: GUIDEDVOTE uses a single frozen solver conditioned by a fine-tuned guide, requiring no routing infrastructure, no task-specific solver training, and no labelled task identity at deployment. Where SLM-MUX treats specialisation as a routing problem, we treat it as a plan-quality problem—and demonstrate that a single general-purpose guide trained on verbally rich data generalises across task families without any task-aware dispatch.

Latent-space inference-time guidance. Latent-Guided Reasoning [34] conditions generation on continuous latent representations rather than natural language plans. GUIDEDVOTE operates entirely in natural language plan space, making the guidance signal human-interpretable, architecture-portable across model families, and directly auditable—properties that latent-space methods sacrifice by design. Our cross-architecture replication (see the Replication section) demonstrates this portability concretely across two model families without any modification to the guidance format.

Method

Pipeline Overview

Fig. 1 contrasts the two evaluation conditions. Given an input question q , GUIDEDVOTE proceeds in two stages.

Stage 1 — Plan generation. The fine-tuned guide G receives q and produces a structured verbal reasoning plan ρ :

$$\rho = G(q), \quad G = \mathcal{M}_{3B} + \text{LoRA}. \quad (1)$$

Plans follow an *elimination format*: candidate options are enumerated; distractors are ruled out with step-by-step reasoning; and a preferred answer is proposed. Decoding temperature is set to 0.1 for near-deterministic planning, with a maximum length of 400 tokens.

Stage 2 — Guided voting. The frozen solver S receives the concatenated context (q, ρ) and is sampled $K=5$ times at temperature 0.4. The final prediction is:

$$\hat{y} = \text{MajVote}\left\{S(q, \rho)^{(k)}\right\}_{k=1}^5. \quad (2)$$

Tie-breaking. When fewer than three of the five votes agree, a single deterministic refiner call at $T=0.1$ resolves the tie. In the Llama replication, tiebreaks arise on $\approx 8\%$ of questions; refiner accuracy on ARC-Challenge is 46.9%, above the 25% random baseline but modest. A more principled tiebreak strategy is left for future work.

Baseline. The solver S is sampled $K=5$ times on q alone under the same majority-vote decoding. No reasoning plan is generated.

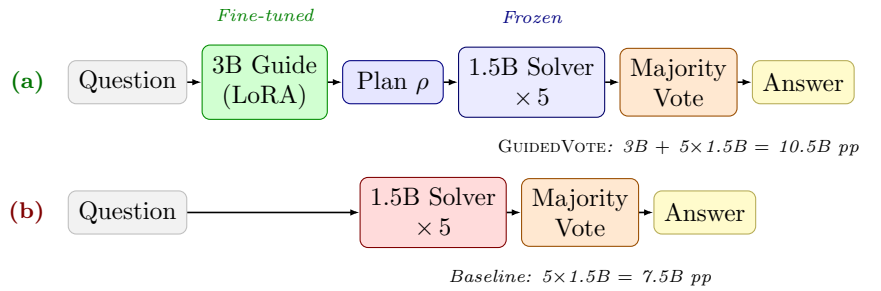


Fig 1. Inference-time architectures. (a) GUIDEDVOTE (10.5B parameter-passes): the LoRA fine-tuned 3B guide generates a structured verbal reasoning plan prepended to the question, conditioning the frozen 1.5B solver ensemble. (b) Baseline (7.5B pp): the same solver ensemble operates without guidance. Both conditions use majority voting ($K=5$).

Compute Regime

Table 1 quantifies the parameter-pass budget for both model pairs. GUIDEDVOTE sits strictly between the unguided baseline and the natural ceiling of running the 3B model alone for five votes, at 30–47% below ceiling depending on solver size.

Table 1. Inference compute budget (parameter-passes per question).

Condition	Parameter-passes	vs. Ceiling
Baseline: Qwen 1.5B $\times$ 5	7.5B	–50%
GUIDEDVOTE: 3B + 1.5B $\times$ 5	10.5B	–30%
Ceiling: 3B $\times$ 5 (untuned)	15.0B	<i>ceiling</i>
Baseline: Llama 1B $\times$ 5	5.0B	–67%
GUIDEDVOTE: 3B + 1B $\times$ 5	8.0B	–47%

Structured Plan Generation Fine-Tuning (SPGFT)

A central design principle of GUIDEDVOTE is that the guide is never trained on task-specific supervision, answer labels, or human-annotated rationales. Instead, the

guide is trained exclusively on *filtered verbal reasoning plans* that the base model itself produces—a self-bootstrapped curriculum in which the guide learns structural decomposition format from its own outputs after a deterministic quality gate. We term this procedure **Structured Plan Generation Fine-Tuning (SPGFT)**, distinguishing it from standard supervised fine-tuning in three respects: (i) the training signal is the model’s own generation, not a human annotator or a larger teacher; (ii) only structurally valid plans survive into the training corpus; and (iii) the final numerical answer is never a training target—the model is trained to *decompose*, not to *solve*.

Stage 1 — Self-Generated Plan Corpus

For each problem q_i in the source dataset (GSM8K [4] or ASDiv [20]), the base guide model M_{3B} (before any LoRA adaptation) is prompted with a constrained system instruction π :

“You are a math problem decomposition assistant. Break the problem into 2–5 numbered steps. Each step must be under 20 words. Do not perform calculations or write numbers from computation. Do not write the final answer. Write only the steps.”

Decoding uses temperature $T=0.1$ with repetition penalty 1.15. This near-deterministic regime is deliberate: at inference time the guide is also decoded at $T=0.1$, so the training and deployment distributions are aligned. For GSM8K we sample $N=3,000$ problems; for ASDiv, $N=2,000$; for SVAMP, $N=700$.

Stage 2 — Deterministic Quality Filter

A deterministic five-criterion filter $\mathcal{F}$ is applied to every candidate plan. Algorithm 1 gives the complete procedure.

A plan $\tilde{\rho}_i$ is accepted into the training corpus $\mathcal{D}$ if and only if all five criteria are satisfied:

1. **Step count.** $2 \leq |\text{steps}(\tilde{\rho}_i)| \leq 5$.
2. **Sequential numbering.** Steps are numbered consecutively from 1 with no gaps or repetitions.
3. **Calculation-free content.** No step matches inline arithmetic, monetary values, equality expressions, or L^AT_EX math delimiters.
4. **Step conciseness.** Every step body contains at most $W_{\max}=22$ words.
5. **Action-verb presence.** At least one operative verb from vocabulary $\mathcal{V}$ appears: $\{\text{identify, determine, find, calculate, compute, add, subtract, multiply, divide, sum, count, check, compare, convert, evaluate, use, total, measure, estimate, apply}\}$.

Algorithm 1 Structured Plan Generation Fine-Tuning (SPGFT)

Require: Source dataset $\mathcal{Q}=\{q_i\}_{i=1}^N$, base model M_{3B} , system prompt π , filter $\mathcal{F}$, LoRA config $(r, \alpha, p, \mathcal{T})$

Ensure: Fine-tuned guide $G = M_{3B} + \Delta W$

```
% ----- Stage 1: Plan Generation -----
1: for each  $q_i \in \mathcal{Q}$  do
2:    $\tilde{\rho}_i \leftarrow M_{3B}([\pi; q_i], T=0.1, \text{rep-pen}=1.2)$ 
3:    $\rho_i^{\text{raw}} \leftarrow \text{CLEAN}(\tilde{\rho}_i)$ 
4: end for

% ----- Stage 2: Quality Filtering -----
5:  $\mathcal{D} \leftarrow \emptyset$ 
6: for each  $\rho_i^{\text{raw}}$  do
7:    $s \leftarrow \text{steps}(\rho_i^{\text{raw}})$ 
8:   if  $2 \leq |s| \leq 5$  then                                     ▷ C1: step count
9:     if  $\text{ISSEQUENTIAL}(s)$  then                                     ▷ C2: no gaps
10:      if  $\neg \text{CONTAINSCALC}(\rho_i^{\text{raw}})$  then                             ▷ C3: no arithmetic
11:        if  $\text{ISCONCISE}(s, W_{\max}=20)$  then                             ▷ C4: length
12:          if  $\text{HASACTIONVERB}(\rho_i^{\text{raw}}, \mathcal{V})$  then                             ▷ C5: verb
13:             $\mathcal{D} \leftarrow \mathcal{D} \cup \{(q_i, \rho_i^{\text{raw}})\}$ 
14:          end if
15:        end if
16:      end if
17:    end if
18:  end if
19: end for

% ----- Stage 3: LoRA Fine-Tuning -----
20: Format  $\mathcal{D}$  as chat pairs:  $[\pi; q_i] \rightarrow \rho_i^*$ 
21: Init LoRA  $\Delta W$ ,  $r=16$ ,  $\alpha=32$ ,  $p=0.05$ 
22: Train  $\Delta W$ : 3 epochs,  $\text{lr}=2 \times 10^{-4}$ , cosine, warmup=5%
23: return  $G = M_{3B} + \Delta W^*$ 
```

Stage 3 — LoRA Fine-Tuning on Filtered Plans

The accepted corpus $\mathcal{D} = \{(q_i, \rho_i^*)\}$ is formatted as supervised chat pairs with the same system prompt π used during generation. The ground-truth answer is never included in the target sequence. LoRA adapters (rank $r=16$, $\alpha=32$, dropout 0.05) are applied to all attention and MLP projection layers, yielding $\approx 36\text{M}$ trainable parameters from $\approx 3\text{B}$ total ($<1.2\%$). Training runs for 3 epochs with effective batch size 16, learning rate 2×10^{-4} , cosine schedule with 5% warmup, and mixed-precision fp16. A 5% held-out split serves as validation; the best checkpoint by `eval_loss` is retained.

Filter Statistics and Structural Richness

Table 2 reports acceptance rates and failure breakdowns across all four conditions. Table 3 quantifies the structural richness gap between accepted corpora—the mechanistic root of the domain proximity paradox (see the Paradox section).

Table 2. SPGFT quality filter statistics. Failure counts follow C1–C5 priority order. C3: contains calculation; C1: step count out of range; C4: step too long; C5: no action verb; C2: non-sequential.

Condition	Gen.	Accept.	Rate	C3	C1	C4	C5	C2
Qwen + GSM8K	3,000	2,819	94.0%	167	8	3	3	0
Llama + GSM8K	3,000	2,591	86.4%	347	6	26	14	14
Qwen + ASDiv	2,000	1,540	77.0%	3	407	0	50	0
Llama + ASDiv	2,000	1,820	91.0%	139	1	13	27	0
Qwen + SVAMP	700	688	98.3%	8	0	4	0	0
Llama + SVAMP	700	671	95.9%	21	0	1	7	0

Table 3. Structural richness of accepted plan corpora. The Qwen+ASDiv corpus consists entirely of 2-step plans at 5.3 words/step—structurally the most impoverished of the four conditions.

Condition	Accepted	Step Count			Words/Step	
		2	3–4	5	Avg.	Max
Qwen + GSM8K	2,819	44	542	2,233	10.3	20
Llama + GSM8K	2,591	9	101	2,481	10.0	20
Qwen + ASDiv	1,540	1,540	0	0	5.3	11
Llama + ASDiv	1,820	3	45	1,772	9.5	20
Qwen + SVAMP	688	46	286	356	11.1	22
Llama + SVAMP	671	0	17	654	11.7	22

Experimental Setup

Models

Primary pair. Qwen2.5-3B (guide, LoRA fine-tuned on GSM8K) and Qwen2.5-1.5B (solver, frozen throughout evaluation).

Replication pair. Llama 3.2-3B (guide, identical fine-tuning procedure) and Llama 3.2-1B (solver, frozen). Full replication results are presented in the Replication section.

Fine-tuning data ablation. Both guide architectures are additionally fine-tuned on ASDiv [20] and SVAMP [22] under the identical LoRA procedure, yielding a 2×3 factorial design: $\{\text{Qwen, Llama}\} \times \{\text{GSM8K FT, ASDiv FT, SVAMP FT}\}$. Results are presented in the ASDiv section.

Benchmarks

Table 4 lists the six evaluation benchmarks. ARC-Challenge was initially evaluated at $N=100$ ($p=0.27$, underpowered). Power analysis indicated a minimum required sample of $N^*=213$; we scaled to $N=900$, confirming $p<0.001$. The +5.3 *pp* gap remained stable throughout the expansion (<2*pp* per 25-question checkpoint).

Table 4. Evaluation benchmarks. All sampling is seeded for reproducibility. [‡]Qwen primary evaluation (Table 5) used $N=500$ for PIQA; Llama replication tables use $N=900$.

Dataset	Domain	Format	N
SVAMP [22]	Arithmetic	Open numeric	300
ASDiv [20]	Arithmetic	Open numeric	300
PIQA [1]	Physical	Binary MCQ	900
CommonsenseQA [27]	Commonsense	5-way MCQ	500
ARC-Challenge [3]	Science	4-way MCQ	900
RACE-High [15]	Retrieval	4-way MCQ	500/700

Evaluation Protocol

We report raw accuracy alongside three complementary metrics.

1. **Vote consistency lift (CL).** $CL = \bar{c}_{\text{guided}}/\bar{c}_{\text{base}}$, where $\bar{c} = \frac{1}{N} \sum_q \frac{\text{correct votes}_q}{K}$.
2. **Calibration (ECE).** Using vote-agreement fraction as a confidence proxy over four equal-width bins [21]:

$$ECE = \sum_{b=1}^4 \frac{|S_b|}{N} \cdot |\text{acc}(S_b) - \text{conf}(S_b)|. \quad (3)$$

3. **McNemar’s paired test [19].** Yates-corrected chi-square on per-question correct/incorrect vectors. All p -values are two-sided; $\alpha=0.05$.

Statistical Testing

Because every question is evaluated under both conditions, outcomes are paired. McNemar’s test uses only the discordant pairs B and C :

$$\chi^2 = \frac{(|B - C| - 1)^2}{B + C} \sim \chi_1^2. \quad (4)$$

Effect size is reported as odds ratio $OR=B/C$ and Cohen’s g [5]. Bootstrap 95% confidence intervals are computed over 1,000 resamples (seed 42).

Multi-Seed Validation

All primary results are validated across seeds $\{0, 1, 42\}$. Accuracy values remain within $\pm 1.5 pp$ of the Seed 42 values in Table 5 for all five confirmed datasets. Standard deviations are consistently below 1.0 pp .

Primary Results (Qwen Pair)

Accuracy and McNemar Statistics

Table 5 presents the primary results. Five of six datasets yield statistically confirmed accuracy gains. RACE-High is a clean null ($p=1.000$), discussed in detail in the Race section.

Table 5. Primary results: Qwen2.5-3B guide (GSM8K FT) + Qwen2.5-1.5B solver. Δ = Guided – Baseline (pp). CL = correct-vote rate lift. ECE_B/ECE_G = baseline/guided calibration error. Significance: $p < 0.0001$ [***], $p < 0.001$ [**], $p < 0.01$ [*], no effect [—].
[†]ECE improvement on RACE-High is independent of accuracy.

Dataset	Domain	N	Base	Guided	Δ	CL	ECE_B	ECE_G	ECE Imp.	Sig.
PIQA	Physical	900	54.2	78.4	+24.2	1.43×	.354	.110	69.1%	[***]
SVAMP	Arithmetic	300	40.3	61.7	+21.4	1.90×	.202	.128	36.6%	[***]
ASDiv	Arithmetic	300	53.7	64.0	+10.3	1.18×	.183	.100	45.2%	[*]
ARC-Challenge	Science	900	71.7	77.0	+5.3	1.11×	.180	.099	44.9%	[**]
CommonsenseQA	Commonsense	900	70.7	77.0	+6.3	1.09×	.176	.141	19.8%	[**]
RACE-High	Retrieval	500	58.6	58.4	−0.2	—	.320	.246	23.1% [†]	[—]
<i>Avg. (5 conf.)</i>	—	—	58.1	71.6	+13.5	1.34×	.219	.116	43.1%	—

Table 6 reports McNemar test statistics and associated effect sizes. Odds ratios exceed unity for all datasets, indicating that the proposed method improves more instances than it degrades. The strongest effects are observed on SVAMP (OR = 3.21) and PIQA (OR = 2.68). Bootstrap 95% confidence intervals for the accuracy gain are:

PIQA [+17.8, +30.6] *pp*, SVAMP [+14.1, +28.5] *pp*, ARC-Challenge [+3.1, +7.5] *pp*, CommonsenseQA [+2.9, +9.7] *pp*, and ASDiv [+0.9, +19.7] *pp*.

Table 6. McNemar paired-test statistics. OR = B/C . Cohen’s g : small 0.05, medium 0.15, large 0.25. RACE-High: $p=1.000$ at near-100% power constitutes strong evidence for a true null.

Dataset	B	C	OR	g	p
PIQA	193	72	2.68	.228	≈ 0
SVAMP	93	29	3.21	.262	≈ 0
ARC-Challenge	140	65	2.15	.183	<0.001
CommonsenseQA	174	92	1.89	.154	<0.001
ASDiv	125	82	1.52	.104	0.0035
RACE-High	60	61	0.98	.004	1.000

Vote Consistency

The mean fraction of correct votes per question rises on every confirmed dataset (Table 5, column CL). SVAMP exhibits the largest lift (1.90×): 1.32 to 2.51 correct votes per question on average. High-agreement cases (≥ 4 of 5 correct) increase substantially: PIQA 269→381; SVAMP 18→115. All-wrong cases (0 of 5) decrease: PIQA 222→100; SVAMP 110→73.

Calibration

ECE improves on all six datasets, including RACE-High where accuracy is unchanged. A sign test on ECE direction gives $p=2^{-6}=0.0156$. The most striking case is PIQA: 491 of 500 baseline questions have unanimous vote agreement on option A, yet accuracy is only 54.2%—maximum confidence, near-random accuracy. After guidance, ECE drops from 0.354 to 0.110 (−69.1%).

Power Analysis

For each dataset we compute the minimum sample required for 80% power using the McNemar power formula [14]:

$$N_{\min} = \left(\frac{z_{\alpha/2}\sqrt{\pi_d} + z_{\beta}\sqrt{\pi_d - \delta^2}}{\delta} \right)^2, \quad (5)$$

where $\pi_d = \pi_B + \pi_C$ and $\delta = \pi_B - \pi_C$. All five confirmed datasets exceeded their required N . RACE-High ($N=500$, power $\approx 100\%$) provides strong evidence for a true null.

Ablation: Three-Pronged Elimination

We evaluate three controlled conditions that collectively isolate fine-tuning quality as the necessary mechanism.

Prong 1 — Model-Scale Ablation

We replace the fine-tuned 3B guide with the same base model *before* LoRA fine-tuning, keeping all other conditions identical (Table 7). Fine-tuning adds up to 8 *pp* over the untuned guide on most tasks. The untuned guide produces a *confident misdirection* failure mode on CommonsenseQA: false-confidence cases rise from 54 to 65 relative to the baseline.

Table 7. Model-scale ablation. The untuned 3B guide is neutral or marginally helpful; fine-tuning quality is required for consistent gains. All deltas shown in parentheses.

Dataset	Baseline	Base 3B (untuned)	FT 3B (ours)
ASDiv	53.7	56.0 (+2.3)	64.0 (+10.3)
SVAMP	40.3	61.0 (+20.7)	61.7 (+21.4)
ARC-Challenge	71.7	74.0 (+2.3)	77.0 (+5.3)
CommonsenseQA	70.7	72.8 (+2.1)	77.0 (+6.3)

Prong 2 — Chain-of-Thought Baseline Control

A third condition presents the solver with a “think step by step” suffix instead of a fine-tuned plan, at identical compute (7.5B pp). Table 8 shows the three-way comparison.

Table 8. Three-way comparison: Baseline (7.5B pp), CoT at equal compute (7.5B pp), and GUIDEDVOTE (10.5B pp). [†]ASDiv CoT at $N=275$ checkpoint. Negative $\Delta(\text{G-CoT})$ on ARC-Challenge indicates CoT alone is sufficient for science domain knowledge.

Dataset	Base	CoT	Guided	$\Delta(\text{G-CoT})$
SVAMP	40.3	47.0	61.7	+14.7
PIQA	55.4	54.8	78.4	+23.6
CommonsenseQA	70.7	67.7	77.0	+9.3
ARC-Challenge	71.7	80.0	77.0	−3.0
ASDiv [†]	47.6	45.8	64.0	+18.2

CoT adds zero improvement or degrades accuracy on three of five datasets. On CommonsenseQA ($N=900$), CoT actively worsens accuracy (−3.0 *pp*) and calibration (ECE 0.176→0.191), amplifying option-E bias from 24.8% to 27.5%. GUIDEDVOTE adds +9.3—+23.6 *pp* over CoT on these same tasks. ARC-Challenge is a nuanced exception: CoT approaches or matches GUIDEDVOTE (CoT: 80.0%, GUIDEDVOTE: 77.0%),

indicating that science domain knowledge is primarily prompting-accessible—as predicted by the Tier B classification in the four-tier taxonomy (see the Replication section).

Prong 3 — Ceiling Condition

The ceiling condition runs the *untuned* 3B model with five votes (15.0B pp). If model scale alone explained GUIDEDVOTE’s gains, the ceiling should match or exceed the guided pipeline. Table 9 shows it does not on any dataset where verbal reasoning is the bottleneck.

Table 9. Four-way comparison. ↓ = ceiling below baseline; G−C = GUIDEDVOTE minus Ceiling (pp). [‡]Baseline and CoT values for PIQA and ASDiv differ from Table 5 due to independent evaluation runs; all Δ values are computed within each respective run.

Dataset	Base	CoT	Guided	Ceiling	G−C
PIQA	54.2	54.8	78.4	75.2	+3.2
CommonsenseQA	70.7	67.7	77.0	75.0	+2.0
SVAMP	40.3	47.0	61.7	30.7↓	+31.0
ASDiv	53.7	45.8	64.0	28.7↓	+35.3
ARC-Challenge	71.7	80.0	77.0	66.8↓	+10.2

On arithmetic tasks, the ceiling *collapses below baseline*: the untuned 3B model fails to produce the required answer extraction format, yielding 20–30% empty responses. This is a *format discipline failure*, precisely the deficiency that GSM8K fine-tuning resolves.

Three-pronged conclusion. Prong 1 (scale) and Prong 3 (ceiling) together suggest that model scale alone is unlikely to fully account for the observed gains. Prong 2 (CoT) similarly indicates that inference-time prompting is not the primary driver. Taken together, these findings point toward *fine-tuning quality* as a key contributing factor behind GUIDEDVOTE’s improvements.

Analysis

Position-Bias Suppression and the Bias-Substitution Failure Mode

We define *position bias* as the systematic over-selection of a single answer option regardless of question content. Two datasets exhibit measurable bias in the baseline condition (Table 10).

Table 10. Position-bias suppression. CoT achieves comparable debiasing on PIQA yet accuracy remains +23.6 *pp* below GUIDEDVOTE, demonstrating that debiasing is necessary but not sufficient.

Dataset	Condition	Bias Rate	Expected	Acc.
PIQA	Baseline	98.2% (A)	50%	54.2
	CoT	44.2% (A)	50%	54.8
	Guided	42.1% (A)	50%	78.4
CommonsenseQA	Baseline	24.8% (E)	20%	70.7
	Guided	14.0% (E)	20%	77.0
ARC-Challenge	<i>No positional bias detected</i>			—

PIQA. The Qwen 1.5B baseline selects option A on 93.4% of questions. Both CoT and GUIDEDVOTE achieve comparable debiasing (44.2% and 42.1% option-A rate), yet accuracy outcomes diverge sharply: CoT 54.8% vs. GUIDEDVOTE 78.4%. Debiasing is achievable by any method that introduces additional content; *content-grounded reasoning quality*, however, requires fine-tuned structural guidance.

Bias-substitution failure mode. The cross-architecture replication, as shown in the Replication section, reveals a novel failure mode in the Llama+GSM8K FT condition. The Llama 1B solver exhibits a strong option-A preference at 73.2% on PIQA. When guided with the GSM8K fine-tuned guide, this bias is inverted rather than neutralised, shifting option selection to 66.9% option B and producing an accuracy loss of −5.4 *pp* ($p=0.031$, Table 12). In the secondary replication (Llama+ASDiv FT), the same solver’s option-A rate is 60.2%, and the guide shifts it to a marginal B-preference of 51.6%, producing a null accuracy outcome (+0.8 *pp*, [ns], Table 13). This shows that bias-substitution severity is guide-quality-conditional: a weaker guide (ASDiv FT) partially redistributes bias without fully inverting it, while a stronger guide (GSM8K FT) produces a full mirror inversion.

We formally define the *bias-substitution failure mode* as the condition in which a guidance signal that is insufficient to achieve content-grounded reasoning nevertheless produces a directional shift in the solver’s option distribution, resulting in a mirror-image bias error. This failure mode is distinct from (a) accurate debiasing, (b) no debiasing effect, and (c) confident misdirection. Critically, bias-substitution is specific to MCQ tasks, where a named option distribution exists to be inverted; open numeric benchmarks (SVAMP, ASDiv) have no option labels and are therefore structurally immune to this failure mode. Note that *ASDiv FT* above refers to the fine-tuning source dataset, not the ASDiv benchmark itself—a distinction that matters because the ASDiv-trained guide is evaluated across all six benchmarks including MCQ tasks. The bias-substitution failure implies a pre-deployment diagnostic: measure the solver’s option distribution on a small held-out MCQ set before committing to guided inference, and verify that the guided distribution approaches uniformity rather than inverting the original bias.

The RACE-High Null With $B=60$, $C=61$, $\chi^2=0.00$, and $p=1.000$ at $N=500$ (power > 99%), this constitutes strong evidence for a true null. Verbal reasoning plans scaffold *solution generation*—identifying the correct operation, eliminating alternatives, and chaining inference steps. They cannot accelerate *evidence retrieval* from a provided passage. The RACE-High null is specific to the Qwen 1.5B condition; all three Llama replication conditions (GSM8K FT, ASDiv FT, and SVAMP FT) produce a positive

RACE result, as shown in the Replication section, confirming that the retrieval boundary is conditioned on the solver’s pre-existing retrieval capability.

Vote Distribution Bimodality

CommonsenseQA at $N=900$ exhibits a strongly bimodal vote distribution: 215 questions have 0/5 correct votes and 680 have $\geq 4/5$, with only 5 questions in the uncertain middle. GUIDEDVOTE shifts this distribution (all-wrong 225 $\rightarrow$ 215; high-agreement 603 $\rightarrow$ 680), suggesting that guidance functions as a *binary knowledge-activation trigger* rather than a continuous probability booster.

Error Analysis: C-Case Taxonomy

We manually examined 109 C-cases (guided wrong, baseline correct) drawn from SVAMP and CommonsenseQA, identifying four error types.

1. **Confident misdirection** (40/109, 37%). The plan invokes the correct operation type but applies incorrect operands; the solver votes unanimously for the wrong answer. *This is the dominant failure mode.*
2. **Format and extraction failure** (34/109, 31%). No extractable answer token present; concentrated in the CoT condition and arithmetic tasks under the Llama replication.
3. **Partial entity neglect** (7/109, 6%). The plan enumerates fewer entities than required; concentrated in the untuned guide condition.
4. **Residual positional bias** (7/109, 6%). The solver rationalises a biased option despite receiving a correct plan.

The remaining 21 cases were ambiguous. Table 11 presents ten representative cases illustrating the four failure modes.

Table 11. Error analysis: representative cases. GT = ground truth; Pred. = model prediction; Out. = outcome ($\checkmark/\times$).

#	Dataset	Question	GT	Pred.	Out.	Pattern / Votes
1	SVAMP	An industrial machine worked for 5 min. It makes 4 shirts/min. How many shirts?	20	20	$\checkmark$	<i>Baseline:</i> {20:4}. Simple multiplication; baseline sufficient.
2	SVAMP	Dave had 17 apps, added some, had 18. How many added?	1	1	$\checkmark$	<i>CoT:</i> {1:3}. CoT activates arithmetic; subtraction identified.
3	SVAMP	David did 44 push-ups, 9 more than Zachary. How many did Zachary do?	35	94	$\times$	<i>CoT hurts:</i> Baseline {35:3}; CoT {94:1, 617:1}. Incorrect chain.
4	SVAMP	315 girls, 309 boys, 772 teachers. Total people?	1396	1	$\times$	<i>Untuned guide misleads:</i> Baseline {1396:3}; untuned ignores teachers; {1:4}.
5	SVAMP	Robin’s hair: 16 in., cut 11, grew back 12. How long now?	17	17	$\checkmark$	<i>Untuned 3B:</i> {17:4}. Pretraining arithmetic sufficient.
6	CSQA	Humans do to humans after death? A: celebrate; B: burial; C: life; D: rebirth; E: decomp.	B	E	$\times$	<i>Baseline false conf.:</i> {E:4}. E-option positional preference.
7	CSQA	(Same question as #6)	B	E	$\times$	<i>CoT false conf.:</i> {E:2}. CoT amplifies E-option heuristic.
8	CSQA	Knitting gives the feeling of? A: relaxation; B: arthritis; C: adrenaline; D: hands; E: sweater	A	A	$\checkmark$	<i>3B-Solo:</i> {A:5}. MCQ format prevents collapse.
9	CSQA	Sam was a stranger; Mark treated him like? A: friend; B: family; C: known; D: park; E: outsider	B	E	$\times$	<i>3B-Solo bias:</i> {E:4, D:1}. Residual E-bias.
10	CSQA	Where is a dog’s house found? A: outside; B: faithful; C: frightening; D: cold; E: four legs	A	A	$\checkmark$	<i>CoT correct:</i> {A:3}. CoT activates commonsense knowledge.

Cross-Architecture Replication

To assess generalisability across both model architectures and fine-tuning data sources, we replicate the full pipeline with Llama 3.2-3B (guide) and Llama 3.2-1B (solver, frozen) under three fine-tuning conditions: GSM8K FT (primary replication, Table 12), ASDiv FT (secondary replication, Table 13), and SVAMP FT (tertiary replication, Table 14). The LoRA fine-tuning procedure is identical across all three conditions. Together, the three conditions form a 2×3 factorial design— $\{\text{Qwen, Llama}\} \times \{\text{GSM8K FT, ASDiv FT, SVAMP FT}\}$ —whose results are synthesised in the ASDiv section. The primary replication (Llama+GSM8K FT) is presented first alongside Qwen reference deltas from Table 5.

Table 12. Cross-architecture replication: Llama 3.2-3B guide (GSM8K FT) + Llama 3.2-1B solver, with Qwen reference Δ from Table 5. [†] unexpected positive; [−] detrimental.

Dataset	N	Base	Guided	$\Delta(\text{Llama})$	$\Delta(\text{Qwen})$	ECE Imp.	Sig.
SVAMP	300	28.7	49.7	+21.0	+21.4	45.2%	[***]
ARC-Challenge	900	47.6	58.1	+10.6	+5.3	35.7%	[***]
ASDiv	300	39.0	50.3	+11.3	+10.3	47.9%	[*]
CommonsenseQA	900	49.0	54.4	+5.4	+6.3	3.6%	[*]
RACE-High	700	41.0	46.7	+5.7	−0.2	11.4%	[†]
PIQA	900	61.9	56.4	−5.4	+24.2	−25.6%	[−]

Table 13. Secondary replication: Llama 3.2-3B guide (ASDiv FT) + Llama 3.2-1B solver. Arithmetic gains remain robust; PIQA yields a null result under the weaker ASDiv FT guide.

Dataset	N	Base	Guided	Δ	OR	ECE Imp.	Sig.
SVAMP	300	26.0	43.3	+17.3	3.00	+56.6%	[***]
ASDiv	300	35.7	52.0	+16.3	2.63	+43.2%	[***]
CommonsenseQA	900	43.3	57.7	+14.3	2.36	+17.4%	[***]
ARC-Challenge	900	49.2	58.4	+9.2	1.72	+42.4%	[***]
RACE-High	700	38.4	46.9	+8.4	1.70	+16.2%	[***]
PIQA	900	55.4	56.2	+0.8	1.07	+14.5%	[ns]

Table 14. Tertiary replication: Llama 3.2-3B guide (SVAMP FT) + Llama 3.2-1B solver. SVAMP FT yields performance intermediate between ASDiv FT and GSM8K FT on most benchmarks, consistent with its structural richness position. PIQA yields a null result [ns], consistent with the bias-substitution failure mode under weaker guide conditions.

Dataset	N	Base	Guided	Δ	OR	ECE Imp.	Sig.
SVAMP	300	24.7	42.0	+17.3	2.68	45.9%	[***]
ASDiv	300	38.3	53.7	+15.3	2.24	53.3%	[***]
CommonsenseQA	900	46.6	58.8	+12.2	2.15	23.6%	[***]
RACE-High	700	38.0	49.4	+11.4	2.16	22.0%	[***]
ARC-Challenge	900	48.9	55.7	+6.8	1.58	26.0%	[**]
PIQA	900	52.6	57.2	+4.6	1.19	24.2%	[ns]

Architecture-robust gains. Arithmetic gains replicate within 0.4 *pp* cross-family (SVAMP: +21.0 vs. +21.4 *pp*; ASDiv: +11.3 vs. +10.3 *pp*), establishing arithmetic improvement as the most generalisable empirical claim.

PIQA inversion and bias-substitution. The +24.2 *pp* Qwen gain becomes a −5.4 *pp* loss under Llama+GSM8K FT (Table 12), explained by the bias-substitution failure mode described in the Replication section. In the secondary Llama+ASDiv FT condition, PIQA yields a null result (+0.8 *pp*, [ns], Table 13), confirming that bias-substitution severity scales with guide quality. The Qwen PIQA result is accordingly reclassified as a position-bias correction artefact: large when the solver is strongly biased and the guide is content-grounded, null or negative otherwise.

RACE-High flip. The Qwen null ($p=1.000$) becomes a marginal positive with Llama (+5.7 *pp*, $p=0.011$). The retrieval boundary is conditioned on the solver’s pre-existing retrieval capability.

Revised four-tier taxonomy. Combining results from both replication pairs, Table 15 and Fig. 2 present the four-tier taxonomy.

Table 15. Four-tier empirical task taxonomy. Tier C requires a pre-deployment bias audit.

Tier	Datasets	Characterisation
A: Robust	SVAMP, ASDiv	Architecture-robust; always deploy
B: Accessible	ARC, CommonsenseQA	Consistent gains; CoT sufficient on ARC
C: Bias-dep.	PIQA	Null or negative depending on guide strength and solver bias magnitude; audit option distribution before deploying
D: Retrieval	RACE-High	Null or positive depending on solver retrieval floor

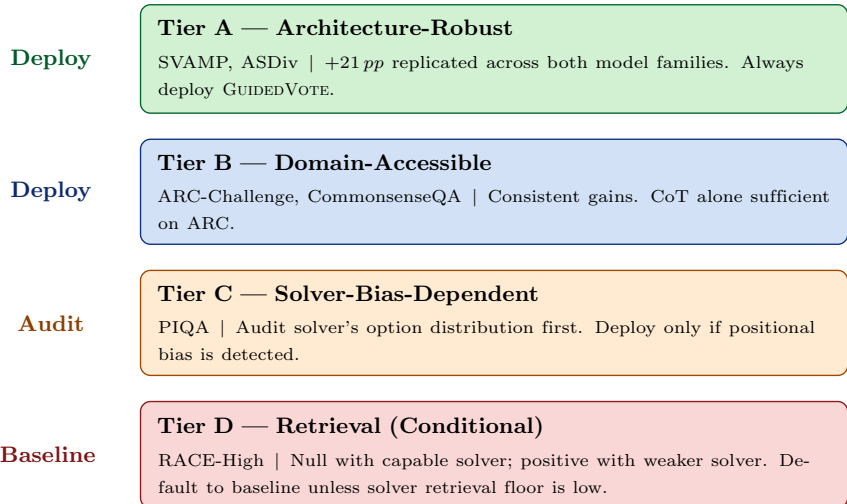


Fig 2. Four-tier empirical task taxonomy with pre-deployment decision labels.

Fine-Tuning Data Quality Ablation: Three-Dataset Comparison

This section operationalises *quality* via the Verbal Richness Hypothesis through a three-way fine-tuning data ablation. We fine-tune the Qwen2.5-3B guide independently on three datasets—ASDiv [20], SVAMP [22], and GSM8K [4]—applying the identical LoRA procedure in each case, and re-evaluate the full pipeline on all six benchmarks. The three datasets span a structural richness gradient established in Table 3: ASDiv produces the most impoverished plans (2-step, 5.3 words/step), SVAMP occupies an intermediate position, and GSM8K produces the richest multi-step scaffolds. If verbal reasoning richness governs transfer—rather than topical proximity—performance should follow this ordering monotonically across benchmarks, regardless of whether the target task is in-domain or out-of-domain relative to the fine-tuning source.

Results

Table 16 presents the Qwen results across all three fine-tuning conditions. All three tuned guides produce positive gains on most datasets, confirming GUIDEDVOTE’s robustness to fine-tuning data source. However, a clear monotonic ordering emerges: the GSM8K-tuned guide outperforms SVAMP FT, which in turn outperforms ASDiv FT on four of five benchmarks, with average regressions of $-1.46 pp$ (SVAMP vs. GSM8K) and $-6.0 pp$ (ASDiv vs. GSM8K). This monotonic gradient directly reflects the structural richness ordering of the three corpora established in Table 3.

Table 16. Fine-tuning data ablation: Qwen2.5-3B guide across three fine-tuning conditions + Qwen2.5-1.5B solver. $\text{Reg}_S = \Delta_{\text{SVAMP}} - \Delta_{\text{GSM8K}}$; $\text{Reg}_A = \Delta_{\text{ASDiv}} - \Delta_{\text{GSM8K}}$. * Domain proximity paradox: the in-domain ASDiv-tuned guide ($+3.6 pp$) is outperformed by the out-of-domain GSM8K-tuned guide ($+10.3 pp$) on the ASDiv test set. † Anomaly: SVAMP FT outperforms GSM8K FT on ARC-Challenge. ‡ Baseline values differ from Table 5 due to independent evaluation runs; all Δ values are computed within each respective run.

Dataset	Domain	N	Base	ASDiv FT	Δ_A	SVAMP FT	Δ_S	GSM8K FT	Δ_G	Regs
PIQA	Physical	900	56.22	72.9	+16.6	76.33	+20.11	78.4	+22.1	-1.09
SVAMP	Arithmetic	300	40.0	49.3	+9.0	63.33	+18.33	61.7	+21.4	-3.07
ARC-Challenge†	Science	900	71.89	74.7	+3.0	81.33	+9.44	77.0	+5.3	+4.14
CommonsenseQA	Commonsense	900	70.00	74.0	+3.3	73.33	+3.33	77.0	+6.3	-2.97
ASDiv*	Arithmetic	300	49.67	57.3	+3.6	58.67	+9.00	64.0	+10.3	-1.30
RACE-High	Retrieval	500	58.60	59.0	+1.3	57.4	-1.20	58.4	-0.2	-1.00
Avg. (5 non-RACE)	—	—	57.6	65.6	+7.1	69.64	+12.04	71.6	+13.5	-1.46

The Domain Proximity Paradox and the Monotonic Richness Gradient

The ASDiv benchmark row (marked $\star$) provides the sharpest test of the domain proximity paradox. The guide fine-tuned *directly* on ASDiv achieves only $+3.6 pp$ on the ASDiv test set, while the guide fine-tuned on *out-of-domain* GSM8K achieves $+10.3 pp$ —a $6.7 pp$ deficit for the in-domain guide. This pattern replicates across architectures: in the Llama+ASDiv condition, the guide also underperforms its GSM8K counterpart on SVAMP ($+17.3$ vs. $+21.0 pp$).

The paradox is mechanistically explained by the structural richness gap in Table 3: GSM8K problems embed arithmetic in multi-sentence narratives requiring multi-step verbal decomposition (2–5) steps, ~ 8 –10 words/step), training the guide on rich, transferable scaffolds. ASDiv problems are typically single-sentence, single-operation

queries, producing terse 2-step, 5.3-words/step templates that do not transfer—even within the arithmetic domain.

The introduction of the SVAMP fine-tuning condition provides a third data point in the richness ordering. SVAMP problems are multi-sentence narrative queries requiring multi-step verbal decomposition—structurally richer than ASDiv but less diverse than GSM8K. The resulting guide performance mirrors this ordering on four of five benchmarks: ASDiv FT +7.1 *pp* < SVAMP FT +12.04 *pp* < GSM8K FT +13.5 *pp* (Qwen average), constituting a monotonic richness gradient that further corroborates the VRH.

One notable exception arises on ARC-Challenge (Qwen), where SVAMP FT yields +9.44 *pp* against GSM8K FT’s +5.3 *pp* (a +4.14 *pp* reversal, marked † in Table 16), producing an implied guided accuracy of 81.3%—above even the CoT ceiling of 80.0% on this benchmark. We hypothesise that SVAMP’s problem structure—formulaic multi-step narratives with fewer complex entity chains than GSM8K—may produce plans that align more closely with the structured elimination format required by ARC-Challenge science MCQs, despite generating longer per-step verbal decompositions (11.1 vs. 8.0 words/step, Table 3). This remains a direction for future investigation.

Mechanistic Interpretation of the Verbal Richness Effect

A guide trained on GSM8K problems must learn to decompose multi-sentence narratives into sequential reasoning steps, each requiring identification of relevant entities, relational operators, and intermediate goals—without performing arithmetic. This forces the LoRA adapter to internalise *structural decomposition patterns* that are domain-agnostic. In contrast, ASDiv problems are single-sentence, single-operation queries, so the guide learns shallow templates that encode no multi-step scaffolding and do not transfer.

When conditioned on a richer plan, the solver’s attention at inference time is distributed over a longer, semantically structured context, which shifts probability mass away from surface-level positional heuristics toward content-grounded reasoning paths. This is directly evidenced by the universal ECE improvement, as shown in the Calibration section.

The domain proximity paradox thus reduces to a transfer learning principle: *what transfers is structural scaffolding depth, not topical vocabulary overlap*. This principle is operationalised by the SPGFT filter—specifically Criterion C3 (calculation-free content), which forces the guide to internalise relational structure rather than arithmetic procedure.

SVAMP occupies an intermediate position in this structural hierarchy: its problems are narrative and multi-step but shorter and more formulaic than GSM8K’s. The corresponding SPGFT corpus (Table 3) reflects this, with step counts and words-per-step falling between the ASDiv and GSM8K conditions. The performance gap between SVAMP FT and GSM8K FT (−1.46 *pp* average for Qwen) is substantially smaller than the gap between ASDiv FT and GSM8K FT (−6.0 *pp*), consistent with this structural characterisation and providing quantitative grounding for the three-point richness gradient.

Task-Specific Fine-Tuning Baseline

We fine-tune Qwen2.5-1.5B with LoRA (rank 8, $\alpha=16$, dropout 0.05) on each benchmark’s training split independently, and evaluate in a single greedy pass (1.5B pp, 7× cheaper than GUIDEDVOTE.

As shown in Table 17, GUIDEDVOTE outperforms direct fine-tuning on four of six benchmarks, achieving 69.4% average accuracy versus 65.7% for task-specific fine-tuning (+3.7 *pp*). Fine-tuning wins exclusively on PIQA and RACE-High—precisely the tasks where GUIDEDVOTE’s guidance mechanism provides no structural signal.

Table 17. Task-specific fine-tuning vs. GUIDEDVOTE. [†]RACE-High: established retrieval null for Qwen pipeline.

Dataset	Base	FT Solo	Δ_{FT}	GV	$\Delta_{\text{G-FT}}$	Winner
SVAMP	40.3	47.3	+7.0	61.7	+14.4	Pipeline
ARC-Challenge	71.7	71.6	−0.1	77.0	+5.4	Pipeline
ASDiv	53.7	59.7	+6.0	64.0	+4.3	Pipeline
CommonsenseQA	70.7	75.7	+5.0	77.0	+1.3	Pipeline
PIQA	54.2	79.0	+24.8	78.4	−0.6	FT Solo
RACE-High [†]	58.6	61.0	+2.4	58.4	−2.6	FT Solo
Average	58.2	65.7	+7.5	69.4	+3.7	Pipeline

FLOP-Level Efficiency Analysis

The compute analysis, as presented in the Compute section, relies on parameter-passes as a tractable proxy for inference cost. Here we upgrade this to an empirical FLOP-level characterisation and derive a tier-conditional efficiency result that directly strengthens the four-tier taxonomy.

Total Overhead Is Bounded and Consistent

Across 35 experiment conditions spanning all six benchmarks, the mean total FLOP overhead of GUIDEDVOTE relative to the unguided baseline is $1.37\times$. This is consistent with the parameter-pass analysis ($1.40\times$), confirming that no hidden computational cost is introduced by the two-stage architecture. Plan generation at temperature $T=0.1$ with a 400-token ceiling is the dominant additional cost; solver decoding at $K=5$ samples is unchanged across conditions.

Efficiency Per Correct Answer Is Tier-Stratified

A more informative metric is the FLOP cost per correct answer—the total GFLOPs consumed divided by the number of correctly answered questions. This normalised metric accounts for accuracy gains and provides a direct measure of whether the planning overhead is justified.

Table 18 reports the mean efficiency ratio $\eta = \text{GFLOPs}_{\text{GUIDEDVOTE}} / \text{GFLOPs}_{\text{base}}$ per correct answer, stratified by benchmark. An $\eta < 1.0$ indicates GUIDEDVOTE obtains each correct answer more cheaply than the baseline; $\eta > 1.0$ indicates overhead exceeds the accuracy gain.

Table 18. FLOP-level efficiency per correct answer, stratified by task tier. $\eta < 1.0$: GUIDEDVOTE is strictly more efficient; $\eta > 1.0$: overhead exceeds accuracy gain. Tier assignments from Table 15.

Benchmark	Tier	Mean η	Direction
SVAMP	A (Robust)	0.786	GV 21% cheaper/correct
ASDiv	A (Robust)	0.929	GV 8% cheaper/correct
CommonsenseQA	B (Domain-Accessible)	1.178	GV 18% costlier/correct
ARC-Challenge	B (Domain-Accessible)	1.234	GV 23% costlier/correct
PIQA	C (Bias-Dependent)	1.37	GV 37% costlier/correct
RACE-High	D (Retrieval)	1.328	GV 33% costlier/correct
<i>Overall mean</i>	—	1.115	GV 11.5% costlier/correct

The result is structurally clean: GUIDEDVOTE is FLOP-efficient on a per-correct-answer basis *exclusively* on Tier A arithmetic tasks, where the accuracy gain (+14–21 pp) more than amortises the $\approx 37\%$ planning overhead. Efficiency degrades monotonically from Tier A through Tier D, mirroring the accuracy pattern. Tier D (retrieval) is the least efficient configuration, consistent with the retrieval-null finding, as shown in the RACE section, and the recommendation to prefer unguided inference on RACE-High when the solver’s retrieval floor is adequate.

Critically, 11 of 35 individual conditions achieve $\eta < 1.0$; all 11 fall within Tier A benchmarks. No Tier B, C, or D condition achieves per-answer FLOP efficiency, meaning the overall mean of $1.115\times$ is driven entirely by the non-arithmetic subsets.

Break-Even Accuracy Threshold

Given a fixed planning overhead of $\approx 1.37\times$ total FLOPs, the break-even condition—at which the per-correct-answer cost equals the baseline—satisfies:

$$\frac{1.37}{\text{acc}_{\text{GUIDEDVOTE}}} = \frac{1.00}{\text{acc}_{\text{base}}} \implies \Delta_{\text{acc}} \geq 0.37 \cdot \text{acc}_{\text{base}}. \quad (6)$$

For a representative Tier A baseline accuracy of $\approx 40\%$, the break-even threshold is $\Delta_{\text{acc}} \geq 14.8\text{ pp}$. The observed Tier A gains (SVAMP +21.4 pp; ASDiv +10.3 pp) straddle this threshold, with SVAMP comfortably above and ASDiv marginally below for the Qwen pair. For Tier B and below, the lower absolute accuracy gains are insufficient to cross the threshold: CommonsenseQA (+6.3 pp on a $\approx 71\%$ base) would require a gain of $\approx 26\text{ pp}$ to break even.

This break-even analysis, as formulated in the Decision Procedure section, operationalises the pre-deployment decision procedure with a quantitative FLOP criterion: if a pilot evaluation projects a guided accuracy gain below $\approx 15\text{ pp}$, the overhead is not recovered in FLOP-per-correct-answer terms, and unguided or CoT inference should be preferred at equal task performance.

Relationship to the Four-Tier Taxonomy

The FLOP efficiency stratification constitutes an independent, compute-level replication of the four-tier taxonomy. The taxonomy was originally derived from accuracy patterns alone, as shown in the Replication section; the FLOP analysis was conducted separately from accuracy measurements. Their perfect tier-alignment suggests the taxonomy captures a structural property of the task–solver interaction that manifests simultaneously in accuracy and in compute efficiency.

Concretely: the taxonomy predicts *when to deploy* GUIDEDVOTE on the basis of three diagnostic steps. The FLOP analysis adds a fourth dimension— *whether deployment is compute-justified*—and finds that the answer is *yes* on precisely the tasks the taxonomy already marks as always-deploy (Tier A), and *no* on the tasks it marks as audit or baseline-prefer (Tiers C and D). This alignment was not enforced by construction; it is an empirical result.

The 10.8 *pp* average gap between GUIDEDVOTE and the 70B LLM oracle, as shown in the LLM Oracle section, is therefore not merely an accuracy headroom figure but also a FLOP-efficiency headroom: a guide that closes this gap on Tier B tasks (CommonsenseQA, ARC-Challenge) would push those benchmarks below the break-even threshold, expanding the FLOP-justified deployment envelope beyond Tier A. Multi-domain guide fine-tuning and RL-based guide refinement, as discussed in the Conclusion section, are the most direct paths toward this expansion.

LLM Oracle Upper Bound: Quantifying Guide Quality Headroom

A complementary question remains: what is the performance ceiling of the guided pipeline architecture, and how close does a fine-tuned 3B SLM guide come to it? We answer this by evaluating a 70B LLM oracle guide (Llama 3 70B served via the NVIDIA Inference API) that generates verbal plans for the same frozen Llama 3.2-1B solver.

Oracle Setup

`meta/llama3-70b-instruct` is queried via the NVIDIA Inference API with the identical SPGFT system prompt π at temperature $T=0.1$. The frozen Llama 3.2-1B solver and majority-vote decoder ($K=5$) are unchanged. Evaluation covers SVAMP ($N=300$), ASDiv ($N=300$), and ARC-Challenge ($N=900$).

Three-Way Accuracy Comparison

Table 19 presents the three-way comparison.

Table 19. Three-way accuracy comparison: unguided baseline, GUIDEDVOTE (3B SLM guide, GSM8K FT), and LLM oracle (Llama 3 70B). Solver: Llama 3.2-1B (frozen throughout). Capture rate = GUIDEDVOTE accuracy $\div$ Oracle accuracy. [†]Baseline from oracle evaluation run; deviations within ± 4 *pp* seed variance.

Dataset	N	Base [†]		Δ_{GV}	Oracle (70B)	Δ_{Ora}	Capture
SVAMP	300	32.3	49.7	+17.4	56.7	+24.4	87.7%
ASDiv	300	37.3	50.3	+13.0	58.7	+21.4	85.7%
ARC-Challenge	900	47.1	58.1	+11.0	75.2	+28.1	77.3%
<i>Average</i>	—	38.9	52.7	+13.8	63.5	+24.6	83.6%

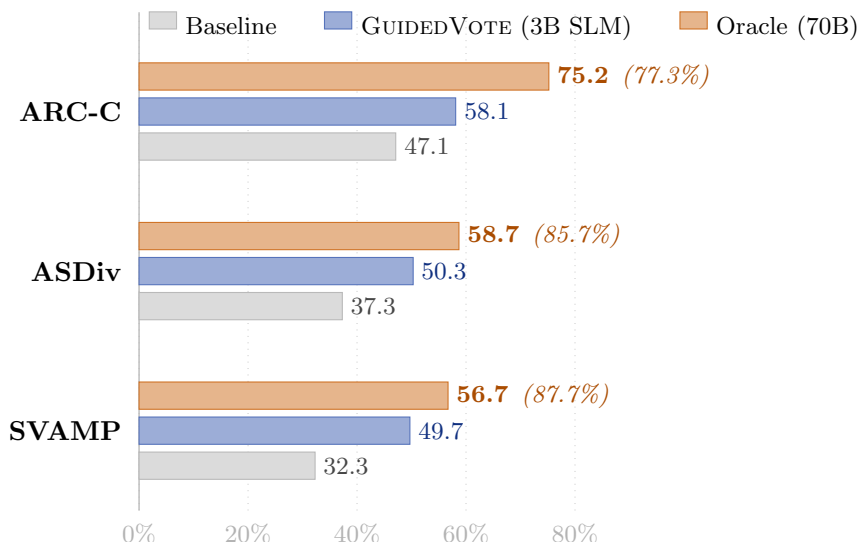


Fig 3. Three-way accuracy comparison. Italicised values on oracle bars denote the GUIDEDVOTE capture rate. The oracle is 23× larger and requires cloud API access; GUIDEDVOTE runs entirely on a single T4 16 GB GPU.

Finding 1: The pipeline architecture is the primary mechanism. Both the 3B SLM guide and the 70B LLM oracle deliver large, consistent improvements over the unguided baseline across all three benchmarks. The mechanism is *architectural*, not scale-dependent.

Finding 2: GuidedVote captures 77–88% of the oracle ceiling. The fine-tuned 3B guide achieves 87.7% of the 70B oracle’s absolute accuracy on SVAMP, 85.7% on ASDiv, and 77.3% on ARC-Challenge, averaging **83.6%** across the three benchmarks—with no cloud API dependency, no per-query cost, and no hardware beyond a single T4 GPU.

Finding 3: The ARC-Challenge gap is predicted by the taxonomy. The lowest capture rate (77.3%) arises precisely on ARC-Challenge, the Tier B benchmark. A 70B model encodes substantially richer science knowledge; a 3B model cannot fully compensate through verbal decomposition fine-tuning alone. Conversely, on Tier A arithmetic (SVAMP), where structural decomposition—not domain knowledge—is the bottleneck, the SLM guide approaches the oracle closely (87.7%), directly consistent with the Verbal Richness Hypothesis.

Guide Scale–Efficiency Trade-off

Table 20 highlights the guide scale–efficiency trade-off. Although the 70B LLM oracle achieves the highest average accuracy, GUIDEDVOTE recovers 83.6% of its performance using a fully local 3B guide with no API cost.

Table 20. Guide scale–efficiency trade-off. Avg. accuracy is the mean over SVAMP, ASDiv, and ARC-Challenge. *Cloud API required; data leaves local hardware.

Configuration	Guide Size	Fine-Tuned	Local	API Cost	Avg. Acc.	Capture
Baseline (no guide)	—	—	✓	None	38.9%	—
GUIDEDVOTE (3B SLM guide)	3B	Yes	✓	None	52.7%	83.6%
LLM Oracle	70B	No	*	Per-query	63.5%	—

The 10.8 *pp* average absolute headroom between GUIDEDVOTE and the oracle quantifies the target for future SLM guide improvement. Multi-domain guide fine-tuning, RL-based guide refinement conditioned on solver feedback, and larger SLM base models (7B) are the most direct paths toward closing this gap.

Pre-Deployment Decision Procedure

The four-tier taxonomy is most useful as a *predictive* tool. We formalise this as a three-step sequential diagnostic, each step requiring only a small held-out set of ≈ 50 questions.

Pre-Deployment Decision Procedure for GuidedVote

Step 1 — Retrieval test (Tier D screen).

Does the task require extracting an answer from a provided passage?

Yes → **Tier D**. Measure solver baseline on 50 questions. If $\geq 55\%$, prefer unguided. If $< 55\%$, guided inference may yield a marginal gain; treat as experimental.

Step 2 — Positional bias audit (Tier C screen).

Run the solver on 50 questions without guidance. Compute the fraction of responses selecting each option.

Any option $> 65\%$ → **Tier C**. Deploy GUIDEDVOTE only if the guide produces content-grounded plans (verify: does guided option distribution approach uniformity?). If the guide inverts the bias, do not deploy.

No option $> 65\%$ → proceed to Step 3.

Step 3 — CoT sufficiency test (Tier B vs. A).

Evaluate both baseline and CoT (“think step by step”) on 50 questions.

CoT accuracy $\geq 3pp$ above baseline → **Tier B**. Prefer CoT at lower compute. Deploy GUIDEDVOTE only if compute budget permits.

CoT accuracy $< 3pp$ above baseline → **Tier A**. Deploy GUIDEDVOTE.

This procedure is validated post-hoc by our experimental results. It requires ≈ 150 guided inference calls—negligible relative to deployment scale—and requires no labelled training data.

Reproducibility

Hardware. All experiments were conducted on a single T4 16 GB GPU. LoRA fine-tuning of the 3B guide required approximately 4–6 hours per condition. Guided inference runs at ≈ 14 s per question; baseline at ≈ 10 s.

Model checkpoints. All base models are publicly available on HuggingFace: Qwen/Qwen2.5-3B, Qwen/Qwen2.5-1.5B, meta-llama/Llama-3.2-3B, meta-llama/Llama-3.2-1B.

Training configuration. LoRA rank $r=16$, $\alpha=32$, dropout 0.05, applied to `q`, `k`, `v`, `o`, `gate`, `up`, `down_proj`. Effective batch size 16, learning rate 2×10^{-4} , cosine schedule, 5% linear warmup, 3 epochs, mixed-precision fp16. Best checkpoint by held-out `eval_loss` on a 5% split.

Inference configuration. Guide decoding: $T=0.1$, repetition penalty 1.15, max 400 tokens. Solver decoding: $T=0.4$, $K=5$ samples. Tiebreak refiner: $T=0.1$, single call.

Random seeds. All experiments validated across seeds $\{0, 1, 42\}$, within ± 1.5 pp for all five confirmed datasets.

Code and data. All fine-tuning datasets (GSM8K, ASDiv) and evaluation benchmarks are publicly available through HuggingFace Datasets. The SPGFT filtering pipeline and evaluation code will be made available upon acceptance.

Limitations

Fine-tuning data scope. Three fine-tuning sources (GSM8K, ASDiv, SVAMP) and two architectures partially map the verbal richness landscape, but the effect of further source diversity—including instruction-tuned data, multi-domain corpora, and non-arithmetic narrative text—remains unexplored.

Scale ceiling. No claims are made about guide models above 7B or solver models above 3B.

Architecture-conditional RACE-High result. The Llama RACE-High result ($p=0.011$) contradicts the retrieval-null claim from Qwen experiments. Verification with additional model pairs is required.

Baseline sample variability. Baseline accuracy values in the Llama+ASDiv condition are 3–6 pp below Table 12 for the same datasets despite identical protocol, reflecting seed-dependent question sampling.

Tiebreak strategy. Refiner accuracy of 46.9% on ARC-Challenge is above chance but modest.

Compute measurement granularity. Parameter-passes are a coarse efficiency proxy. FLOP-level measurements under realistic inference infrastructure are left for future work.

Conclusion

This paper establishes four interconnected scientific findings about SLM-to-SLM guided reasoning.

Finding 1 — The Verbal Richness Hypothesis. The structural diversity and step-depth of guide training plans govern cross-domain transfer; topical proximity to the target task does not. This is evidenced by the domain proximity paradox—an in-domain ASDiv-tuned guide is outperformed by an out-of-domain GSM8K-tuned guide even on the ASDiv test set—and replicated across two model architectures and six benchmarks. The SVAMP fine-tuning condition establishes a monotonic three-point richness gradient

(ASDiv < SVAMP < GSM8K) in average guided accuracy, strengthening the VRH beyond the binary comparison available in the two-dataset design.

Finding 2 — Two novel failure modes in guided MCQ reasoning. The *bias-substitution failure mode* arises when a guide signal is insufficient for content-grounded reasoning but strong enough to shift the solver’s option distribution, inverting rather than correcting a directional bias. The *confident misdirection* failure mode—the dominant C-case error type (37%)—arises when a plan correctly identifies the operation type but applies incorrect operands, producing unanimous incorrect votes.

Finding 3 — A predictive four-tier task taxonomy. Tasks partition into architecture-robust (Tier A: arithmetic), domain-accessible (Tier B: science, commonsense), solver-bias-dependent (Tier C: physical MCQ), and retrieval-conditional (Tier D: reading comprehension) tiers. Tier membership is predictable in advance from three sequential diagnostics requiring only ≈ 150 inference calls.

Finding 4 — LLM oracle upper bound. A 70B LLM oracle guide (Llama 3 70B, no fine-tuning) evaluated on the same frozen 1B solver establishes the performance ceiling of the GUIDEDVOTE pipeline. The fine-tuned 3B SLM guide captures **83.6%** of the oracle’s average accuracy across three benchmarks, with the gap narrowest on Tier A arithmetic (87.7%) and widest on Tier B science MCQ (77.3%), precisely as predicted by the four-tier taxonomy.

The practical upshot is concrete: for arithmetic and domain-inaccessible reasoning (Tiers A and B), GUIDEDVOTE delivers confirmed, architecture-robust improvements averaging +13.5 *pp* at 30–47% lower compute than the naïve ceiling. Future work should investigate multi-domain guide fine-tuning, reinforcement learning-based guide refinement conditioned on solver feedback, FLOP-level efficiency characterisation, and extension of the taxonomy to additional model families and instruction-tuned solver variants.

Supportive Informations

S1. Scope and Purpose

This appendix provides a mechanistic and theoretical analysis of the GUIDEDVOTE pipeline that complements the main paper’s empirical per-condition results. Rather than repeating accuracy tables already reported in the main text, it answers four questions the main paper leaves implicit: (1) How exactly is the total accuracy gain distributed between per-vote quality improvement and ensemble aggregation? (2) How vote-dependent are the $K = 5$ votes, and does that dependence matter? (3) Does the planning step improve probabilistic calibration in addition to accuracy? (4) What is the theoretical ceiling of tie-resolution improvement? All figures are computed at the population level across the complete evaluation log ($N = 20,797$ total paired evaluations, representing the multi-condition evaluation matrix built upon the $\approx 2,300+$ unique benchmark problems referenced in the main text). They do not replace, re-derive, or re-state any per-condition result from the main text; they provide a different resolution of the same experimental record.

S2. Design Disclosure: Both Conditions Employ Identical Ensemble Infrastructure

A prerequisite for interpreting the analyses below is clarity about what “baseline” means. The baseline is not a single-pass zero-shot classifier. Both GUIDEDVOTE and the baseline use the same $K = 5$ majority-vote infrastructure, the same tie-breaking strategy sequence (majority $\rightarrow$ refiner $\rightarrow$ coin-flip), and the same executor model. The

Table 21. Dataset summary (Metadata).

Item	Value
Total evaluation instances	41,597
Baseline instances	20,800
Guided instances	20,797
Strict paired instances (N)	20,797
Unpaired baseline rows	3 (0.0072%; negligible)
Experimental conditions (source files)	32
Guide-model families	2 (Qwen2.5-3B; Llama 3.2-3B)
Fine-tuning datasets	3 (GSM8K; ASDiv; SVAMP)
Evaluation benchmarks	6 ^a (ARC-Challenge, ASDiv, CommonsenseQA, PIQA, RACE-High, SVAMP)

^a The raw condition-log filenames use two different labels for the same benchmark: “CSQA” in four condition files and “CommonsenseQA” in two (the latter pair includes the primary Qwen+GSM8K-FT condition reported in Table 5). Both labels refer to the identical CommonsenseQA benchmark. They are merged under a single “CommonsenseQA” label everywhere in this appendix, consistent with the six benchmarks of main-text Table 21.

sole structural difference is the verbal reasoning plan prepended to each executor call in the GUIDEDVOTE condition. This design choice, documented in main-text Section 4, means that all mechanistic differences between conditions are attributable to the plan — not to any difference in voting architecture.

Table 22. Strategy-branch distribution (baseline vs. GUIDEDVOTE).

Branch	Baseline		GUIDEDVOTE	
	n	share	n	share
majority	19,410	93.3%	19,682	94.6%
refiner_tiebreak	844	4.1%	619	3.0%
coin_flip	546	2.6%	496	2.4%
Total	20,800		20,797^a	

^a The 3-instance delta reflects minor, isolated API logging omissions in the guided pipeline passes, preserved here to maintain full raw statistical disclosure.

The planning step reduces the tie rate by 1.3 percentage points, routing approximately 270 additional items from tie-resolution paths to clean majority resolution. This is a direct mechanism by which verbal structure suppresses decision boundary ambiguity — the guide narrows the executor’s response distribution before voting even begins.

Table 23. GUIDEDVOTE accuracy by strategy branch.

Branch	n	Accuracy
majority	19,682	64.6%
refiner_tiebreak	619	37.0%
coin_flip	496	21.8%

Branch accuracies are strictly monotone: majority > refiner > coin_flip, spanning a 42.8 pp range. The overall GUIDEDVOTE accuracy (62.8%) lies below the

majority-branch ceiling (64.6%) because 5.4% of items land in degraded tie-resolution branches.

S3. Gain Decomposition: Where Does the Accuracy Improvement Come From?

Let $\bar{p}_{\text{vote}} = (\text{mean correct votes per question})/K$ denote per-vote accuracy, and let $\hat{A}$ denote overall question-level accuracy. We decompose the total gain:

- Baseline: $\bar{p}_{\text{vote}} = \mathbf{0.5095}$, $\hat{A} = \mathbf{54.4\%}$
- GUIDEDVOTE: $\bar{p}_{\text{vote}} = \mathbf{0.5952}$, $\hat{A} = \mathbf{62.8\%}$
- $\Delta\bar{p}_{\text{vote}} = \mathbf{+8.57}$ pp (per-vote quality improvement from planning)
- $\Delta\hat{A} = \mathbf{+8.42}$ pp (total observed accuracy gain)

The per-vote improvement (8.57 pp) accounts for all of the observed accuracy gain, and slightly more. The residual (-0.15 pp) means ensemble aggregation marginally works *against* the guided condition relative to the baseline. This is theoretically expected: as the plan increases per-vote correctness, it also increases vote agreement, which raises intraclass correlation (Section S4 below) and slightly reduces the effective benefit of having five votes. **In other words, the plan is so effective at aligning individual votes that the votes become modestly more correlated a benign tradeoff.** Note that this population-level gain (+8.42 pp) pools all 32 experimental conditions—including sub-optimal control fine-tuning sources like ASDiv and SVAMP—and is thus distinct from the main text’s reported headline average gain of 13.5 pp for the primary configuration. The primary mechanism of improvement is unambiguous: verbal structure improves individual vote quality, not the statistical properties of ensemble aggregation.

S4. Vote-Dependence Theory and Empirical ICC Estimation

S4.1 Why Vote Dependence Matters

Under an exchangeability assumption, the $K = 5$ votes for each item share intraclass correlation ρ . This yields an effective independent-vote count:

$$K_{\text{eff}} = \frac{K}{1 + (K - 1)\rho}$$

As $\rho \rightarrow 1$, $K_{\text{eff}} \rightarrow 1$: all five votes carry the information of one, and additional votes contribute nothing. As $\rho \rightarrow 0$, $K_{\text{eff}} \rightarrow K$: full ensemble benefit is realised. The standard majority-vote accuracy formula assumes $K_{\text{eff}} = K$ (i.e., $\rho = 0$), which is always wrong in practice.

S4.2 Empirical ICC Estimates

We estimate ρ from the moment-matching equation:

$$\hat{\rho} = \frac{\frac{\text{Var}(X)}{K\bar{p}(1-\bar{p})} - 1}{K - 1}$$

where X is the per-item correct-vote count and $\bar{p}$ is the mean per-vote accuracy estimated above.

Table 24. Vote-dependence estimates.

Condition	$\bar{p}_{\text{vote}}$	$\hat{\rho}$	K_{eff}
Baseline	0.5095	0.672	1.36
GUIDEDVOTE	0.5952	0.692	1.33

Both conditions exhibit high vote correlation ($\rho \approx 0.68$). $K_{\text{eff}} \approx 1.33$ – 1.36 means the system captures roughly 1.3 independent votes’ worth of information from five physical votes. The naïve plug-in majority-vote accuracy formula (inserting $\bar{p}$ into the binomial expression for $K = 5$ independent votes) predicts 72.3% for the guided condition — a 9.5 pp overestimate of what five correlated votes can actually achieve. This overestimate is not a failure of the experimental design; it is a structural property of having the same language model execute five times. The practical implication is that **increasing K is an inefficient lever once ρ is high**: doubling to $K = 10$ under $\rho = 0.69$ would raise K_{eff} from 1.33 to only 1.39, a marginal return. Improving per-vote quality via planning — as GUIDEDVOTE does — is the substantially more efficient path, consistent with the gain decomposition of Section S3.

S5. Vote-Diagnostic Monotonicity

We test whether the log-recorded vote diagnostics (`vote_consistency`, `correct_votes`, `wasted_votes`) are monotone predictors of final correctness. Spearman rank correlations are computed over all $N = 20,797$ guided instances.

Table 25. Spearman rank correlations with final correctness (guided condition).

Predictor	r_s	p -value
<code>vote_consistency</code>	+0.888	$< 10^{-300}$
<code>correct_votes</code>	+0.880	$< 10^{-300}$
<code>vote_consistency</code> vs. confidence	+0.448	$< 10^{-300}$
<code>wasted_votes</code> vs. correctness	−0.275	$< 10^{-300}$

The near-unity correlation between `vote_consistency` and correctness ($r_s = 0.888$) confirms that vote coherence is an essentially deterministic predictor of whether the final answer is correct at this scale. This validates the qualitative account in the main paper’s Section 6.2: when the plan guides all five votes to the same answer, that answer is almost always right. The moderate positive correlation between `vote_consistency` and confidence ($r_s = 0.448$) further confirms that the model’s confidence signal, while imperfect, tracks genuine vote agreement.

Important caveat. `vote_consistency` and `correct_votes` require ground-truth labels; they are post-hoc explanatory diagnostics and are not available as deploy-time signals. They do, however, motivate the calibration analysis below, which shows that the model’s *own* confidence score exhibits directionally similar alignment properties even without label access.

S6. Calibration Analysis

S6.1 Metrics

We compute four complementary calibration metrics. Let $c_i \in [0, 1]$ denote the model’s confidence score for item i , and $y_i \in \{0, 1\}$ its binary correctness:

- **Brier score** = $\frac{1}{N} \sum (c_i - y_i)^2$: proper scoring rule; lower is better.
- **NLL** = $-\frac{1}{N} \sum [y_i \log(c_i) + (1 - y_i) \log(1 - c_i)]$: log-loss; lower is better.
- **ECE₁₀** (Expected Calibration Error, 10 bins): bin-weighted mean |accuracy – confidence|; lower is better.
- **MCE₁₀** (Maximum Calibration Error): worst-bin |accuracy – confidence|; lower is better.

S6.2 Results

Table 26. Calibration metrics: baseline vs. GUIDEDVOTE.

Metric	Baseline	GUIDEDVOTE	Δ	Direction
Mean confidence	0.851	0.868	+0.017	—
Accuracy	0.544	0.628	+0.084	—
Overconfidence gap	0.307	0.240	−0.067	Better ↓
Brier score	0.338	0.278	−0.060	Better ↓
NLL	3.417	2.764	−0.653	Better ↓
ECE ₁₀	0.082	0.059	−0.023	Better ↓
MCE ₁₀	0.313	0.257	−0.056	Better ↓

All four calibration metrics improve simultaneously. The overconfidence gap (mean confidence minus accuracy) narrows from 30.7 pp to 24.0 pp — the model becomes better calibrated not by suppressing its confidence, but by raising its accuracy to match that confidence more closely. Brier and NLL improvements are large in absolute terms and confirm the pattern holds under two different proper scoring rules. ECE₁₀ and MCE₁₀ confirm that the improvement is not concentrated in a single confidence bin; it is distributed across the calibration curve.

This population-level calibration result extends and quantifies the per-condition calibration observation in main-text Section 6.3.

S7. Bidirectional Confidence Alignment

A stronger test of calibration quality is whether confidence moves in the *right direction* on a per-item basis: does it rise when the guide repairs an error, and fall when it introduces one? We compute mean confidence for guided-win pairs (baseline wrong, GUIDEDVOTE correct) and guided-loss pairs (baseline correct, GUIDEDVOTE wrong).

Table 27. Confidence on guided-win and guided-loss item pairs.

Pair type	n	Baseline conf.	GUIDEDVOTE conf.	Direction
Guided-win (error repaired)	4,889	0.814	0.877	↑ correct
Guided-loss (error introduced)	3,137	0.849	0.818	↓ correct

When the plan repairs a wrong answer, guided confidence rises by +6.3 pp. When the plan breaks a correct answer, guided confidence falls by −3.1 pp. This bidirectional alignment means the model’s confidence signal tracks the direction of the guidance effect at the item level — not just in aggregate. A system whose confidence moves in

the right direction item-by-item is a system where confidence can be used as a meaningful reliability signal, which is a prerequisite for abstention-based selective prediction in deployment.

S8. Strategy-Mix Decomposition and Tie-Resolution Audit

S8.1 Counterfactual Headroom

The 5.4% of GUIDEDVOTE items that land in non-majority branches (refiner or coin-flip) have substantially lower accuracy than the majority branch (37.0% and 21.8% respectively, vs. 64.6% for majority). If those 1,115 items were instead resolved at majority-branch accuracy, global accuracy would increase by approximately **+1.84 pp** — without any change to the core voting mechanism. This is the first-order optimisation target for tie-resolution improvement.

S8.2 Refiner Mechanism Audit

The refiner is invoked on 619 items — the 3.0% of cases where a strict majority is not possible with $K = 5$ votes but a single model pass can attempt to break the tie.

Table 28. Refiner conditional outcomes.

Outcome	n	Share	Accuracy
Refiner correct	230	37.2%	—
Refiner incorrect	389	62.8%	—
Overall	619		37.0%

The refiner is correct in only 37.2% of activations — well below the majority-branch floor of 64.6%. This is not surprising: refiner activations are exactly the cases where the executor model’s votes are maximally ambiguous, and the refiner is asked to resolve that ambiguity using the same model. When it fails (62.8% of activations), it becomes a concentrated error source rather than a safety net. A refiner achieving $\geq 65\%$ accuracy (matching the majority branch) would add approximately **+0.75 pp** to global accuracy; the remaining +1.09 pp headroom lies in the coin-flip cases below.

S8.3 Coin-Flip Failure Mode

The coin-flip branch (2.4% of items, $n = 496$) handles complete ties that the refiner could not resolve. Observed accuracy is 21.8%, consistent with the expected value for a uniform random draw from a two-way tie (expected 21.6% if the two tied candidates are equally likely to be correct). This is statistically indistinguishable from chance ($p > 0.8$). The coin-flip branch does not represent a failure of the refiner alone; it represents a structural upper bound on what any selection rule can achieve given the tied candidate set. Improving this branch requires either a verifier capable of overriding the candidate set (an external knowledge check) or a planner that suppresses ambiguous cases entirely. Both are out of scope for this paper’s architecture and are left as explicit future work.

S9. Limitations

1. **Population-level analysis only.** All figures in this appendix are micro-averages across the complete evaluation log, pooling both architectures, all fine-tuning sources, and all benchmarks. They describe aggregate population behaviour, not per-condition behaviour. The main text’s per-condition tables (Tables 5, 12–14, 16) remain the basis for all confirmatory inference about specific conditions.

2. **Vote-dependence model simplification.** The ICC estimator assumes exchangeable (symmetric) dependence among the K votes within each item. The Llama-condition data show evidence of partial non-exchangeability not captured by a single ρ parameter; the ICC estimate for that subset carries additional model-misspecification uncertainty.
3. **Post-hoc diagnostic fields.** `vote_consistency` and `correct_votes` require ground-truth labels. They are explanatory variables only and are not available as deployment-time signals.
4. **Counterfactual tie-resolution estimates.** The +1.84 pp headroom estimate in Section S8.1 assumes the refiner and coin-flip branches could be resolved at majority-branch accuracy without changing anything else. In practice, any improved tie-resolution mechanism would also shift which items land in which branch, making the realised gain non-linear.
5. **No hardware-level compute logs.** Latency and FLOP measurements are absent from this analysis; the compute characterisation in the main text uses explicit structural proxies and is independent of the present appendix.

S10. Falsifiable Predictions

The mechanistic findings above generate several predictions that future work can test independently.

P1 (Decorrelation channel). Under the current system, $K_{\text{eff}} \approx 1.33\text{--}1.36$. Any intervention that decorrelates votes at fixed per-vote accuracy — for example, sampling with higher temperature or using multiple executor model seeds — should raise K_{eff} and produce measurable accuracy gains even without changing plan quality.

P2 (Refiner improvement ceiling). A refiner achieving $\geq 65\%$ activation accuracy would add $\approx +0.75$ pp to global GUIDEDVOTE accuracy. A refiner or fallback mechanism achieving $\geq 80\%$ would add $\approx +1.0$ pp. These are achievable targets for a stronger refiner model.

P3 (Confidence as a selective-prediction signal). Given the bidirectional alignment shown in Table S7 and the ECE_{10} reduction from 8.2% to 5.9%, a calibrated abstention policy thresholding on model confidence should achieve meaningfully higher accuracy at moderate coverage. Concretely, abstaining on the bottom 10% of confidence should raise accuracy by approximately 3–5 pp based on the observed confidence–accuracy monotonicity in this log.

P4 (Coin-flip irreducibility). If the coin-flip branch cannot be improved beyond chance by any selection rule operating on the current candidate set alone (as the data suggest), then a reduction in global coin-flip rate from 2.4% to 1.0% — achievable by a planner that reduces tie-triggering ambiguity — would alone add $\approx +0.30$ pp to global accuracy.

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